

Module 9.1 — LEGO Collection Catalog: Usability Refinements

1. Purpose and Scope

An extension of [Module 9](#), written after living with the module. It changes **no table, no column and no domain rule**: the three tables stay three tables, values stay hand-maintained, ownership keeps three statuses, and storage stays a flat **area** + **container** list.

Everything here is about the surface — what the grid shows, how it is narrowed and ordered, which flows lose data, and where the collection can be read from outside the application. Where a request would have cost a table, it was answered with a derived field or a filter instead.

Module 9 remains the specification of record. This document only states the deltas.

Actors

Unchanged (M9).

User Stories

US01. One Visual Language for Status

As a user, I want condition, build state, retirement and completeness to look like the same kind of control, so a row reads at a glance instead of mixing coloured text with grey pills.

US02. Ordering Without a Cartesian Menu

As a user, I want to pick *what* to sort by from a short list and flip the direction with one button, instead of scrolling a menu that repeats every field twice.

US03. Value of Gifts

As a user, I want a set I was given to show something more useful than a dash, since I can still see what it is worth against its original retail price.

US04. "Everything in the Garage"

As a user, I want to filter by storage **area** without naming a container, so "everything in the garage" and "everything in the house" are one click.

US05. Retirement Visible in the List

As a user, I want to see that a set is retired from the grid, the way I already see that a copy is incomplete — without opening it.

US06. Both Sides of Every Filter

As a user, I want "only complete" as readily as "only incomplete", and "still on sale" as readily as "retired".

US07. Numbers on the Collection Tab

As a user, I want the totals of whatever I am currently filtering — copies, sets, pieces, cost, value — without switching to the overview.

US08. Forms That Do Not Argue With Me

As a user, I want the example text in a field to be visibly fainter than what I type, so I never mistake a suggestion for a value.

US09. Storage Entry Without Typos

As a user, I want existing areas and containers suggested while I type a new location, so «Garagem» never becomes «garagem».

US10. Creating While Viewing "All Entities"

As a user, I want to create a storage location while the selector is on «todas» by naming the owner in the dialog — the same answer the set dialog already gives.

US11. Editing the Right Copy

As a user, opening a set from the grouped view and editing a copy, I want to choose *which* copy I am editing instead of always landing on the first one.

US12. Removing a Set

As a user, I want a way to delete the catalog entry too, not just its copies, when I registered something by mistake.

US13. Set Years in the Grid

As a user, I want the release and retirement year in the table rather than the purchase date, because that is what tells me whether a set is still buyable.

US14. Take the Collection With Me

As a user, I want one spreadsheet with everything — copies, sets and storage — so I can pivot it, mail it or keep it as a backup.

US15. Evolution Over Time

As a user, I want to see how the collection grew and what it cost, month by month.

US16. Entity Colours in Forms

As a user, I want the entity picker inside a dialog to carry the same colour dot as the global selector.

US17. One Way to Register a Copy

As a user, I want «adicionar outra cópia» to be the same form as «adicionar conjunto», so registering a second copy never silently drops the bank-statement link.

Data Model

No schema change. No table is added, altered or dropped, and no migration is required by this module.

New Derived Fields

Computed on read, persisted nowhere.

Field	On	Definition
<code>rrp_appreciation_eur</code>	<code>LegoSetModel</code>	<code>current_value_eur - rrp_eur</code> ; NULL if either is missing
<code>rrp_roi_pct</code>	<code>LegoSetModel</code>	$(\text{current_value_eur} - \text{rrp_eur}) / \text{rrp_eur} * 100$; NULL if <code>rrp_eur</code> is NULL or 0
<code>summary</code>	GET <code>/lego/instances</code>	<code>copies</code> , <code>unique_sets</code> , <code>total_cost_eur</code> , <code>total_value_eur</code> , <code>total_pieces</code> over every row matching the filters, not the page
<code>timeline</code>	GET <code>/lego/overview</code>	cumulative <code>copies</code> / <code>cost_eur</code> / <code>value_eur</code> per acquisition month
<code>copies_without_date</code>	GET <code>/lego/overview</code>	copies with no <code>acquisition_date</code> , excluded from <code>timeline</code>

Functional Requirements

- **FR-9.14 Workbook Export.** GET `/lego/export.xlsx` returns the whole in-scope collection as one workbook with three sheets — `Cópias`, `Conjuntos`, `Arrumação` — carrying every field the UI can show plus both ROI readings. Read-only, authenticated, entity-scoped like every other read. Requires `openpyxl`; the writer is imported on demand so the dependency never affects application start-up.
- **FR-9.15 Sort Contract.** GET `/lego/instances` takes `sort` (one of `created`, `name`, `pieces`, `year`, `cost`, `value`) and `direction` (`asc` | `desc`) separately. The legacy combined values (`name_desc`, `value_asc`, ...) are still accepted so bookmarked URLs keep working. NULLs sort last in both directions — absent data is not "the smallest".
- **FR-9.16 Hierarchical Storage Filter.** GET `/lego/instances` takes `storage_area`, matching every container in that area. `StorageLocation` remains a **flat** table with no parent/child column: only the filter and the picker are hierarchical.
- **FR-9.17 Tri-state Discovery Filters.** `incomplete_only` and `retired_only` are replaced by `completeness` (`all` | `complete` | `incomplete`) and `retirement` (`all` | `retired` | `available`). Both default to `all`.
- **FR-9.18 Filtered Summary.** GET `/lego/instances` returns a `summary` block aggregated over the whole filtered result set.
- **FR-9.19 Acquisition Timeline.** GET `/lego/overview` returns a cumulative monthly `timeline`. `value_eur` is **today's** value of what was owned by that month, not a historical quote — see [ADR-0009](#).
- **FR-9.20 RRP Reading.** `LegoSetModelOut` exposes `rrp_appreciation_eur` and `rrp_roi_pct`. They never enter any aggregate, KPI or total — see [ADR-0010](#).

- **FR-9.21 Model Deletion from the UI.** Deleting the last copy of a set offers to delete the catalog entry with it, soft or hard, honouring the existing guard that a model with live copies cannot be deleted.
- **FR-9.22 One Copy-Creation Flow.** There is a single create dialog. Registering an extra copy locks the catalog identity and keeps every copy field, including the bank-statement link, box/instructions and missing parts.

UI / Screens

UX-9.8 One Badge Language

Every status badge shares one shape, weight and border; only the tint differs.

- **Condition** is an ordinal quality scale, so colour carries meaning: **Novo** success · **Bom** neutral · **Usado** warning · **Danificado** destructive.
- **Build state** is a neutral fact — sealed is not "better" than built — so it keeps a single neutral tint. Colour is never the only carrier: the label always says it.
- **Incomplete** and **retired** are labelled badges in the set cell. Retirement also tints the whole row, because it is a durable property of the row rather than a transient state.

UX-9.9 Sorting

A field **Select** (Adicionado · Nome · Peças · Ano · Custo · Valor) plus an icon button that flips **asc/desc**. Cost and Valor are retained beyond the requested four because FR-9.13 requires sorting by acquisition cost and current value.

UX-9.10 Collection Columns

Conjunto (thumbnail, name, incomplete/retired badges, set number, piece count) · **Tema** · **Ano** (release › retired) · **Arrumação** · **Estado** · **Condição** · **Custo** · **PVP** · **Valor** · **ROI**. The purchase date leaves the grid and stays in the detail sheet. The ROI cell falls back to the RRP reading, badged **PVP**, when there is no cost basis.

UX-9.11 Filter Bar

Tema · Arrumação (grouped by area, with a "toda a área" entry per area) · Estado · Condição · Propriedade · Completude (tri-state) · Retirados (tri-state).

UX-9.12 Collection Summary Strip

Above the grid: copies, unique sets, pieces, cost and current value for the **current filters**, with an explicit "totais dos filtros aplicados" label so the numbers are never mistaken for the collection totals on the overview.

UX-9.13 Overview Evolution Chart

A line chart above the theme breakdown: cumulative copies (right axis) and cumulative cost plus current value (left axis, EUR). The value line is dashed and the card says in words that it is today's value. Copies without an acquisition date are reported under the chart, never bucketed.

UX-9.14 Forms

- Placeholders use a dedicated `--placeholder` token, visibly lighter than `--muted-foreground`, in inputs, textareas and select triggers alike.
- The entity picker inside dialogs carries the same colour dot as the global selector, and appears in **both** the set dialog and the storage-location dialog whenever the selector is on «todas».
- New storage locations get `data`list suggestions: every existing area, and the containers already used in the area being typed.

UX-9.15 Detail Sheet

- The "Cópias" list is clickable; selecting a sibling switches the sheet to it.
- The edit tab carries a "Cópia a editar" selector whenever the set has more than one copy.
- The set section shows `PVP original`, `Valorização vs PVP` and `ROI vs PVP`.
- The delete dialog offers "eliminar também o conjunto do catálogo" when this is the last copy.
- "Adicionar outra cópia" opens the full create dialog with the set locked.

UX-9.16 Export

An `Exportar` action in the page header downloads the workbook through the API client, so a 401 still routes to the normal re-login flow instead of rendering an error page in a new tab.

Analytics & KPIs

Unchanged. `timeline` and `summary` are new *views* of the same in-collection definition; `rrp_roi_pct` enters no aggregate.

Edge Cases & Business Rules

- **Gifts.** Cost ROI stays `NULL` — the rule does not move. Only the presentation falls back to the RRP reading, and says so.
- **Sets without RRP.** No second reading. Nothing is estimated.
- **Copies without an acquisition date.** Excluded from the timeline and counted separately; never bucketed into "today" or "the earliest month".
- **Legacy sort values.** `?sort=value_desc` still orders by value descending.
- **Area filter and soft-deleted locations.** The area filter resolves through live locations only.
- **Entity on «todas».** Both create dialogs ask for the entity inline. The API still refuses to guess ([ADR-0007](#)).
- **Deleting a model with copies.** Still blocked. The UI only offers it when the last copy is going with it.

Deferred (explicitly not built now)

- A user-configurable column chooser or density toggle.
- An RRP-based KPI or aggregate anywhere on the overview.
- A true market-value history (would need the snapshot table 0008 refused).
- Storage hierarchy in the schema — the filter is hierarchical, the table is not.
- CSV/PDF export, scheduled exports, or export of the audit log.

Definition of Done

- ☐ No migration: `alembic heads` is unchanged by this module.

- ☐ Condition, build state, incomplete and retired render as one badge family; retirement also tints the row and is never signalled by colour alone.
- ☐ Sort is field + direction; **pieces_asc**, **year_asc** and every legacy combined value all resolve correctly; **NULLs** sort last both ways.
- ☐ **storage_area** returns every copy in the area across containers; the schema is untouched.
- ☐ **completeness** and **retirement** are tri-state and default to **all**; the overview's "retired sets" card deep-links to **retirement=retired**.
- ☐ **summary** reflects the filters, not the page.
- ☐ **rrp_appreciation_eur** / **rrp_roi_pct** are derived, absent from every aggregate, and shown with a **PVP** badge when they stand in for cost ROI.
- ☐ **timeline** is cumulative, excludes departed copies, excludes undated copies and reports their count.
- ☐ The workbook exports copies, sets and storage with both ROI readings, and the writer import cannot break application start-up.
- ☐ The storage dialog offers area/container suggestions and an inline entity picker; creating a location while on «todas» works.
- ☐ A second copy is reachable and editable from the grouped view.
- ☐ The last copy can be deleted together with its model, soft or hard.
- ☐ There is exactly one create-copy dialog, and it includes the transaction picker.
- ☐ Placeholders are lighter than muted text everywhere, including select triggers.
- ☐ Tests cover the sort contract (including legacy values), the area filter, the tri-state filters, the filtered summary, the RRP derivation and the timeline.

Open Questions / Decisions

#	Question	Decision	ADR
1	Two ROIs, or a PVP column, or both?	Cost ROI stays the single headline; the RRP reading is derived on the model, shown as a labelled fallback and as a plain PVP column, and enters no aggregate.	0010
2	A time series, when M9 says there is none?	Copies and cost are real history from acquisition_date . The value line is today's value of what was owned by each month — dashed, labelled, and not a price history.	0009
3	Should "só completos" / "só não retirados" exist, or is it overkill?	Neither add a checkbox nor refuse: the two existing booleans become two tri-state selects. Same control count, both directions available, no illegal combination expressible.	—
4	Does hierarchical storage break the flat-table decision?	No. Only the filter and the picker group by area; StorageLocation keeps no parent column and no tree.	—
5	Should build state get colours?	No. Colour is reserved for ordinal judgement (condition). Build state is a neutral fact and keeps one tint — but it now shares the badge shape, which was the real complaint.	—

#	Question	Decision	ADR
6	Why is it wrong to add a location while on «todas»?	It is not wrong, the dialog was incomplete. The API is right to refuse a guessed owner; the fix is to ask, exactly as the set dialog does.	0007
7	Keep the light "add copy" dialog for speed?	No. It silently dropped the transaction link, box/instructions and missing parts. One dialog, with the identity locked when the set is known.	—

Integration Contract

- **Exposes:** one new read endpoint (**GET /lego/export.xlsx**). Still nothing to other modules; the collection's value still does not roll into M6 net worth.
- **Consumes:** unchanged (**Entity, Transaction, Document, Setting, AuditLog**).
- **Guarantees:** no schema change; every new number is derived on read; the RRP reading never contaminates a cost-based aggregate; the export is read-only and entity-scoped like every other read.